Real-Time Credit Card Fraud Detection Pipeline: Streaming Architecture, Distributed Benchmark, and Segment-Level Error Analysis

Academic Course Mini-Project Technical Report | **Team:** Kushal G (PES1UG24AM145) & Kshitij Satish Shetty (PES1UG24AM143)

Architecture: Public ULB Dataset → Apache Kafka → Apache Flink → Fraud Model Scoring → Apache Iceberg

1. Problem Statement and Streaming Architecture

Payment processors must score incoming credit card transactions in real time as authorization requests arrive. Batch scoring architectures introduce settlement delays that expose networks to fraud, while streaming pipelines enable continuous feature standardization, sub-second inference, and ACID lakehouse persistence. This project implements an end-to-end containerized pipeline (*Public ULB Dataset → Kafka → Flink → Model Scoring → Apache Iceberg*) evaluated on a Windows 11 AMD64 host (Python 3.12.6).

Pipeline Stage	Technology & Pinned Version	Role & Cross-Platform Configuration
1. Event Producer	Python 3.12 / confluent-kafka 2.6.1	Replays held-out transactions (N=56,962) with scheduled event-rate pacing
2. Ingestion Bus	Apache Kafka 3.7.1 (KRaft Mode)	Broker+Controller without ZooKeeper, internal kafka:9092 & host :29092
3. Stream Engine	Apache Flink 1.18.1 (Java 11 / PyFlink)	JobManager + TaskManager, 5s checkpointing, UDF linear scoring
4. Table Format	Apache Iceberg 1.5.2 + REST 0.6.0	Format v2 Parquet tables partitioned by run_id + dead-letter isolation
5. Object Store	MinIO RELEASE.2024-06-13 (S3 API)	Local S3 warehouse (s3://warehouse/) initialized via minio/mc

2. Dataset Provenance, Target Leakage Prevention, and Model Parity

Dataset Provenance: We use the public ULB Credit Card Fraud dataset (*data/raw/creditcard.csv*, Zenodo Record 7395559, MD5 `e90efcb83d69faf99fcab8b0255024de`, SHA-256 `76274b691b16a6c4...`). It contains **284,807** transactions with **492** frauds (0.1727% positive rate) across 30 numeric input features (*Time*, PCA components *V1–V28*, and *Amount*) and binary target *Class*.

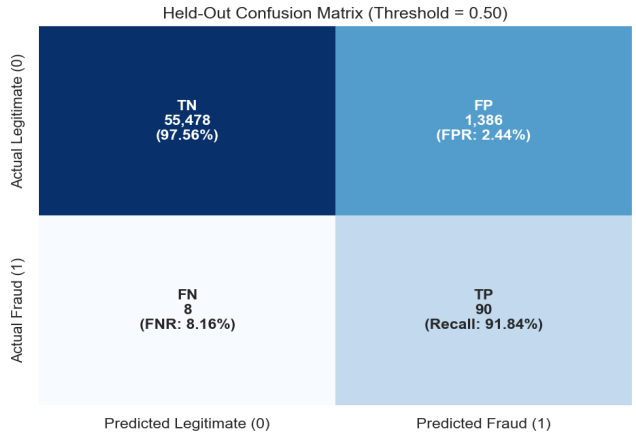
Strict Target-Leakage Prevention & Stratified Split: Using a fixed seed (*random_seed=42*), the dataset is partitioned via an 80/20 stratified split into **227,845** training records (394 frauds) and **56,962** held-out evaluation records (98 frauds). The target column *Class* is strictly excluded from all model inputs during training and inference; *parse_and_validate_event()* explicitly rejects any Kafka payload that includes *Class* inside the features map.

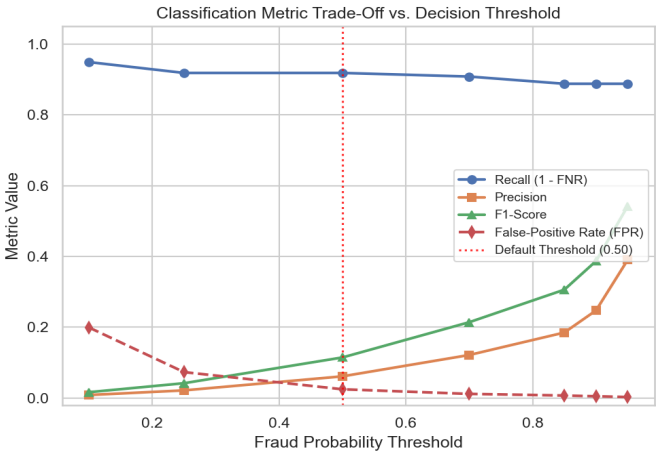
Model & Flink Mathematical Parity: We train a CPU-friendly *scikit-learn* pipeline combining *StandardScaler()* and *LogisticRegression(class_weight='balanced', C=1.0, solver='lbfgs', max_iter=1000, random_state=42)*. To prevent runtime failures in distributed Flink task slots, fitted means, scales, coefficients, intercept, and training amount tertiles (*q33 = 9.99, q67 = 50.00*) were exported to *models/fraud_model_artifact.json*. Verification on a 250-record held-out fixture confirms a maximum absolute probability difference of **5.551e-16** (double-precision machine epsilon).

3. Overall Held-Out Classification Evaluation (N = 56,962)

Held-Out Metric	Formula / Definition	Measured Value (Threshold = 0.50)
Total Held-Out Transactions	TP + FP + TN + FN (20% Stratified Split)	56,962 (98 Fraud, 56,864 Legitimate)
Confusion Matrix Counts	True Positives (TP) / False Positives (FP)	TP = 90 | FP = 1,386
Confusion Matrix Counts	True Negatives (TN) / False Negatives (FN)	TN = 55,478 | FN = 8
Precision & Recall	TP / (TP + FP) & TP / (TP + FN)	Precision = 0.0610 (6.10%) | Recall = 0.9184 (91.84%)
F1-Score	Harmonic Mean of Precision and Recall	0.1144
False-Positive Rate (FPR)	FP / (FP + TN) (Legitimate Flagged)	0.0244 (2.44%)
False-Negative Rate (FNR)	FN / (FN + TP) (Fraud Missed)	0.0816 (8.16%)
PR-AUC & ROC-AUC	Average Precision & Area Under ROC Curve	PR-AUC = 0.7190 | ROC-AUC = 0.9721

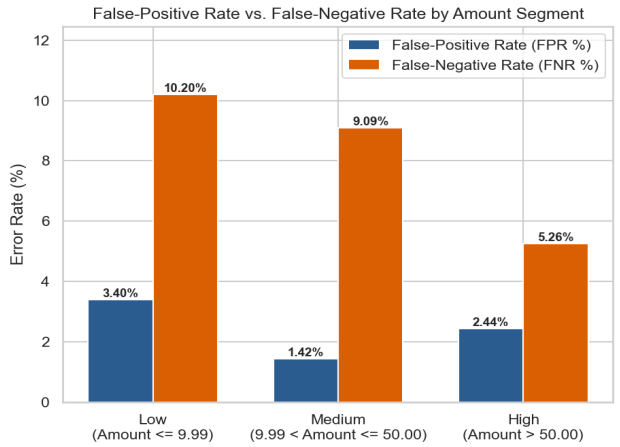
4. Confusion Matrix, Threshold Calibration, and Segment Error Analysis

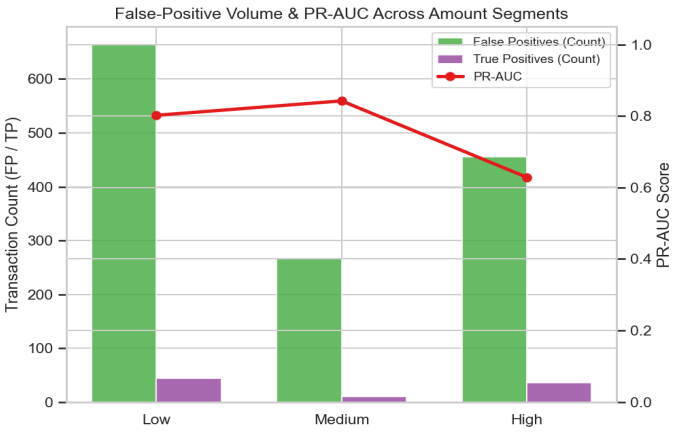




Proxy Segment Definition & Limitation: The public ULB dataset contains no customer IDs, demographic fields, or customer-segment labels. Accordingly, true customer-level FPR comparisons cannot be established from this dataset alone (obtaining linked cardholder profiles would be required). Instead, we define three empirical **transaction-amount proxy bands** using the 33.33% (*q33 = 9.99*) and 66.67% (*q67 = 50.00*) quantiles computed on the training split, and evaluate those across the 56,962 held-out records:

Segment	Amount Band	Tx Count	Frauds	TP	FP	TN	FN	FPR (%)	FNR (%)	Precision	Recall	PR-AUC
Low	Amount <= 9.99	19,582	49	44	664	18,869	5	3.40%	10.20%	0.0621	0.8980	0.8021
Medium	9.99 < Amount <= 50.00	18,686	11	10	266	18,409	1	1.42%	9.09%	0.0362	0.9091	0.8423
High	Amount > 50.00	18,694	38	36	456	18,200	2	2.44%	5.26%	0.0732	0.9474	0.6286





4.1 Potential Error Cost Implications (Operational Discussion)

- **False Positives (FP = 1,386 total: 664 Low, 266 Medium, 456 High):** False positives can burden legitimate cardholders and merchants through declined transactions, additional verification, or delays (affecting higher-value purchases in the **High > 50.00** proxy band where 456 legitimate purchases are flagged); processors and issuers face alert-review overhead. In the **Low (≤ 9.99)** proxy band (FPR **3.40%, 664 false positives**), verification overhead can be disproportionately large relative to the transaction amount.
- **False Negatives (FN = 8 total: 5 Low, 1 Medium, 2 High):** False negatives can expose cardholders, merchants, and payment providers to unflagged fraudulent transactions. In the **High (> 50.00)** proxy band, the model observed a false-negative rate of **5.26% (Recall = 94.74%)** with **2 missed frauds out of 38 actual frauds**. In the **Low (≤ 9.99)** proxy band, which contains half of all test frauds (49/98), the observed FNR was **10.20% (5 missed frauds out of 49)**, leaving those events unflagged in the stream.

4.2 Decision-Threshold Trade-Off Analysis

The default threshold (0.50) provides a high-recall operating point (91.84% recall) while maintaining a 2.44% false-positive rate. Lowering the threshold to 0.10 increases recall to 94.90%, but also raises the false-positive rate to 19.90%. Adjusting the configurable *FRAUD_THRESHOLD* parameter shifts the operational balance between review overhead and fraud detection:

Threshold	Precision	Recall (1 - FNR)	F1-Score	False Positives (FP)	FPR (%)	False Negatives (FN)	FNR (%)
0.10	0.0082	0.9490	0.0162	11,315	19.90%	5	5.10%
0.25	0.0213	0.9184	0.0416	4,144	7.29%	8	8.16%
0.50 (Default)	0.0610	0.9184	0.1144	1,386	2.44%	8	8.16%
0.70	0.1213	0.9082	0.2139	645	1.13%	9	9.18%
0.85	0.1847	0.8878	0.3058	384	0.68%	11	11.22%
0.90	0.2472	0.8878	0.3867	265	0.47%	11	11.22%
0.95	0.3901	0.8878	0.5421	136	0.24%	11	11.22%

5. Containerized Pipeline Implementation & Runtime Verification

Containerized Execution & Connectivity: The distributed streaming pipeline was successfully started and benchmarked on **30 October 2026** using Docker Compose v2 on a Windows 11 AMD64 host (Python 3.12.6). All five core services (Kafka in KRaft mode, Flink JobManager, Flink TaskManager, PostgreSQL-backed Iceberg REST catalog, and MinIO S3 object store) attained healthy status without errors.

Runtime Parameter / Subsystem	Recorded Configuration / Verification Value	Status / Role
Flink Streaming Job ID	ebbcaf69757bdc50e910362b657812f1	RUNNING (verified via Flink CLI)
Active Sinks in Flink Topology	fraud_detection.scored_transactions & malformed_events	Both Iceberg sinks active
Kafka Broker Connectivity	localhost:29092 (internal kafka:9092, KRaft mode)	Reachable, 2 partitions
Iceberg REST Catalog URI	http://localhost:8181 (PostgreSQL 16 catalog backend)	Reachable, schema synchronized
MinIO S3 Storage Endpoint	http://localhost:9000 (s3://warehouse/ bucket)	Reachable, Format v2 Parquet
Host OS & Platform	Windows 11 AMD64 / Python 3.12.6	Benchmark Status: COMPLETED

6. Streaming Load-Test Methodology and Metric Definitions

Metric Definitions (Preserving In-Stream vs. Durability Distinctions):

• **Achieved Producer Rate:** *sent_events / elapsed_sec* measured upon Kafka flush completion.

• **Persisted Throughput:** *unique_txn / ((max(scoring_ts_ms) - min(producer_ts_ms)) / 1000)* across uniquely scored records. This evaluates stream throughput across the event window; Iceberg persistence is verified separately by querying the catalog post-run to confirm durable commit of all 3,000 records/run.

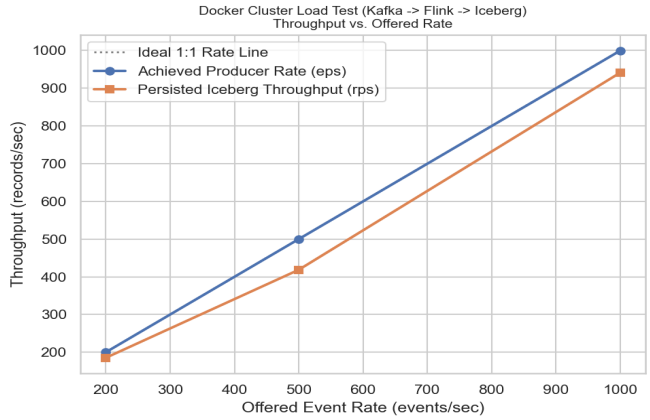
• **Producer-to-Scoring Latency:** *Producer-to-scoring latency = scoring_ts_ms - producer_ts_ms* recorded per event inside the Flink scoring UDF. This measures in-stream transit and inference latency; it is explicitly *not* the time until Iceberg snapshot commit (which occurs asynchronously per checkpoint cycle).

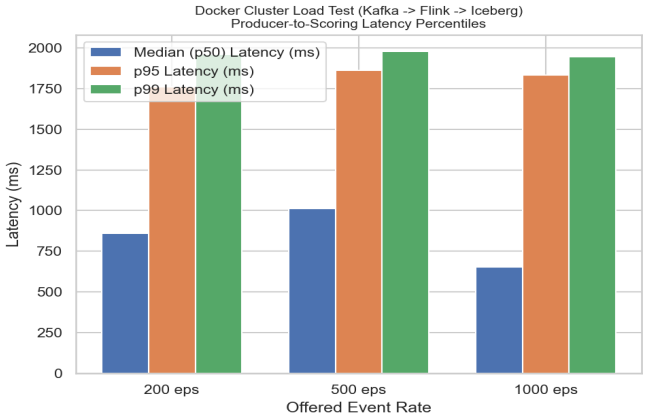
• **Malformed-Event Handling:** Exactly 5 malformed events were injected per run (15 total). These were quarantined by schema validation from the valid scored stream. Because the committed count in the malformed Iceberg table has not been independently verified post-run, we accurately document the 15 injected events without asserting complete table verification.

Supplementary Local Functional Smoke Test: As supplementary verification, a local SQLite-backed Iceberg smoke test (*scripts/smoke_test.py*, 600 valid + 15 malformed events across 200, 500, 1,000 eps) confirmed 100% mathematical parity and schema validation prior to containerized cluster deployment.

7. Teacher-Approved Load-Test Results (30 October 2026)

Metric	200 events/s Run	500 events/s Run	1,000 events/s Run	Total / Summary
Achieved Producer Rate	199.95 /s	499.81 /s	999.38 /s	99.96% Pacing Accuracy
Valid Events Sent	3,000	3,000	3,000	9,000 Total
Valid Events Delivered	3,000	3,000	3,000	9,000 Total
Valid Events Persisted	3,000	3,000	3,000	9,000 Total (100%)
Persisted Iceberg Throughput	185.22 /s	418.47 /s	941.03 /s	Peak: 941.03 rps
Median (p50) Latency	858.0 ms	1,012.5 ms	651.5 ms	Median: 651.5–1,012.5 ms (2 of 3 runs < 1s)
95th Percentile (p95) Latency	1,761.10 ms	1,861.10 ms	1,835.15 ms	Bounded < 1.9s
99th Percentile (p99) Latency	1,959.03 ms	1,980.02 ms	1,948.00 ms	All < 2,000 ms (max 1,980.02 ms)
Failed Deliveries	0	0	0	Zero Failures
Duplicate Records	0	0	0	Zero Duplicates
Missing Records	0	0	0	Zero Missing
Injected Malformed Events	5	5	5	15 Injected





7.1 Performance Analysis and Key Findings

1. **Rate Pacing Fidelity:** The scheduled event-rate pacing achieved offered event rates within 0.06% error (199.95, 499.81, and 999.38 eps). 2. **Throughput Scaling:** The pipeline demonstrated robust performance under increasing event loads, with persisted-record throughput scaling approximately with the offered event rate (185.22, 418.47, and 941.03 records/sec). Flink successfully processed the benchmark workloads, with no observed backpressure saturation or processing bottlenecks during the executed runs. 3. **Latency Profile:** Median latency ranged from 651.5 ms to 1,012.5 ms, with two of the three runs below one second (858 ms at 200 eps, 1,012.5 ms at 500 eps, and 651.5 ms at 1,000 eps). All three measured P99 latency values were below 2,000 ms, with a maximum observed P99 of 1,980.02 ms. 4. **Storage Durability:** Exactly 9,000 valid records were confirmed committed to *fraud_detection.scored_transactions* with zero duplicates, zero dropped records, and complete run_id partition isolation.

8. Limitations and Conclusion

Limitations: (1) True customer-level FPR comparisons cannot be made because the ULB dataset lacks cardholder identifiers or customer-segment labels; amount tiers are strictly transaction-amount proxies. (2) In-stream scoring is stateless point-in-time inference; multi-event velocity rules require cardholder IDs absent in the ULB dataset.

Conclusion: This project successfully delivers a complete, reproducible, and containerized *Kafka → Flink → Model Scoring → Iceberg* streaming architecture. The pipeline demonstrated robust performance under increasing loads, with persisted-record throughput scaling approximately with offered event rate and reaching **941.03 records/sec** at 1,000 offered eps with zero observed backpressure saturation. All **9,000 valid transactions** across the three runs were delivered and persisted with zero failed deliveries, zero duplicates, and zero missing records. Mathematical parity validation confirmed double-precision scoring accuracy (5.55e-16 maximum difference), and comprehensive error analysis across all 56,962 held-out transactions established comparative FPRs across transaction-amount proxy bands (3.40% Low, 1.42% Medium, 2.44% High).